In the figure, our geoshape is represented by the hexagon. Plots A and B are *within* the boundaries of this shape, plot C intersects the hexagon, and plot D is outside the boundaries of the geoshape. Elasticsearch provides the `geo_shape` query to search these plots of various shapes in an envelope built by coordinates.

The `geo_shape` query retrieves locations or addresses represented as shapes in another shape, such as the hexagon in figure 12.6. We construct this hexagonal shape by providing the coordinates as values to an `envelope` field.

Our `cafes` index contains a couple of documents with point-based shapes, so let's write a query to see how to retrieve this data. (If you are curious, index a few more documents with other shapes, too.) The query in the following listing searches for all cafes within the shape defined by the `envelope` field with longitude and latitude pairs.

Listing 12.10 Searching for cafes within a given shape

```
GET cafes/_search
{
  "query": {
    "geo_shape": {
      "address": {
        "shape": {
          "type": "envelope",
          "coordinates": [
            [0.1,55],
            [1,45]
          ]
        },
        "relation": "within"
      }
    }
  }
}
```

Defines a geo_shape query that expects a field and a shape

Sets the shape attribute type that expects an envelope built using coordinates

Defines the relationship between the envelope and the resulting geoshape

The query fetches the documents (cafes) that fall within the envelope constructed by the given pair of longitude and latitude values. In this case, the search returns the Costa Coffee cafe found in the envelope.

One last thing we need to understand is the `relation` attribute. The `relation` attribute defines the relationship of the documents to be found for the given shape. The default value of `relation` is `intersects`, which means the query returns documents that intersect the given shape. Table 12.2 describes the possible values for the `relation` attribute.

Table 12.2 Relationships between documents and the envelope shape

Value of `relation`	Description
`intersects` (default)	Returns documents that intersect the given geometrical shape
`within`	Matches documents if they exist within the boundaries of the given geometrical shape